

Informe de experimento: k3_control_single_call

Generado el 2026-09-16 13:16 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio k3_control_single_call (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f ]
↪ low-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: single_call:structured
- Casos: case14
- Requests: generadas, N=5 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=2, parameterized=1, multistep=1, ambiguous=1
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 3; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 13:14
- Costo total: 0.0697 USD (5 items; costo medio 0.01395 USD/item)
- Items totales: 5; ok=5; con error=0; tokens totales=23144
- Solved global: 20.0% (1/5)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método single_call:structured

System prompt —1697 caracteres; strategy=structured; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.
```

```
## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

```
## Single-call mode
You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to
↪ call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
↪ and do not fabricate numbers.
```

User prompt (primer item) —3819 caracteres; case=case14, seed=1911784258, k=3, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (compact, full case)

Case case14: 14 buses, 15 lines, 5 transformers, total load 243.5 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
↪ index, transformers use 100000+row index.

Buses

```
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.02 MW 12.3 Mvar | gen 44.9 MW vm 1.045 pu
bus 3 | 135 kV | load 68.78 MW 13.87 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 41.1 MW -3.354 Mvar
bus 5 | 135 kV | load 5.462 MW 1.15 Mvar
bus 6 | 0.208 kV | load 12.82 MW 8.586 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 37.69 MW 21.21 Mvar
bus 10 | 0.208 kV | load 11.14 MW 7.182 Mvar
bus 11 | 0.208 kV | load 3.273 MW 1.683 Mvar
bus 12 | 0.208 kV | load 7.419 MW 1.946 Mvar
... (2 líneas omitidas de 14)
```

Branches

```
line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
... (8 líneas omitidas de 20)
```

Available tools

- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Answer by emitting tool calls (function calling) in THIS single response:

- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
↪ non-converged result, say so.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion`
utilidad de tarea Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
single_call:structured	case14	5	20.0 (1/5)	4.30e- 03	1.57 (1/5)	20.0 (1/5)	100.0 (5/5)	4.85e- 03	100.0	-	1.0	4629	-	0.0 (0/5)	100.0 (5/5)	0.0697

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
single_call:structured	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
single_call:structured	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)

Tipos de error de formulación (conteo de items)

Method	missed_step	ok
single_call:structured	4	1

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método `single_call:structured`

Ejemplo: resuelto (solved=True)

```
request (case14-plain-000-s0, case14, difficulty=plain, seed=1813382119): Load case14 and run the power flow, then report the bus with the
↳ lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 239.96871582254337,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: I loaded case14 successfully: 14 buses, 5 generators, 20 lines, 11 loads, total load 239.96871582254337 MW.

I cannot report the bus with the lowest voltage magnitude from the available tool outputs, because no power flow result was returned and no
↳ bus-voltage data is available. A power-flow run is ...[35 chars más]
--- Respuesta final ---
I loaded case14 successfully: 14 buses, 5 generators, 20 lines, 11 loads, total load 239.96871582254337 MW.

I cannot report the bus with the lowest voltage magnitude from the available tool outputs, because no power flow result was returned and no
↳ bus-voltage data is available. A power-flow run is still needed to determine that bus.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4525;
↳ cost_usd=0.0126; wall=5.0 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 243.53734721774015,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
```

```

    gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
Case **case14** was loaded successfully.

2. Key numbers
- Buses: **14**
- Generators: **5**
- Lines: **20**
- Loads: **11**
- Total load: **243.53734721774015 MW**
- Total generation capacity: **772.4000000000001 MW**

3. Violations, if any
No power-flow or v ...[285 chars más]
--- Respuesta final ---
1. One-sentence summary
Case **case14** was loaded successfully.

2. Key numbers
- Buses: **14**
- Generators: **5**
- Lines: **20**
- Loads: **11**
- Total load: **243.53734721774015 MW**
- Total generation capacity: **772.4000000000001 MW**

3. Violations, if any
No power-flow or violation results are available from the tool outputs above.

4. Suggested next step
The requested line disconnection **between bus 10 and bus 11** was **not executed in the available tool outputs**; the next step is to run
↳ the disconnect-line action for buses **10-11** and re-solve.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=3.00e-03; F_MAE=1.62; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=4598; cost_usd=0.0138; wall=4.1 s

```

Ejemplo: no resuelto

```

request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 255.70494989942833,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: I only have one tool output: `load_case("case14")` succeeded.

```

I do not have tool outputs for the required steps:

- disconnect line 6-13
- disconnect line 12-13
- rerun power flow
- list lines with loading above 100%

So I cannot report the overloaded lines without inventing results.

```

1. One-sentence summary
--- Respuesta final ---
I only have one tool output: `load_case("case14")` succeeded.

```

I do not have tool outputs for the required steps:

- disconnect line 6-13
- disconnect line 12-13
- rerun power flow
- list lines with loading above 100%

So I cannot report the overloaded lines without inventing results.

```

1. One-sentence summary
The 14-bus case was loaded successfully, but the requested switching and power-flow results were not executed in the available tool outputs.

2. Key numbers
- Buses: 14
- Generators: 5
- Lines/branches: 20
- Loads: 11
- Total load: 255.70494989942833 MW
- Total generation capacity: 772.4000000000001 MW

3. Violations, if any
Unknown from the available outputs; no post-contingency power-flow or branch-loading results were provided.

```

```

4. Suggested next step
Run the mis ...[213 chars más]
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=1.85e-02; F_MAE=6.25; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=4781; cost_usd=0.0155; wall=5.2 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- V_MAE $\geq 1e-4$ p.u. en 1 filas método×caso con solver: single_call:structured/case14 (4.30e-03).
- El gate no verificó ningún resultado de flujo (gate_checked=0 en todos los items).
- Ningún item terminó con **error**.
- Solved de single_call:structured: 20.0 % (1/5).
- Formulación exacta agregada sobre métodos con tools: 20.0 % (1/5).
- Formulación por dificultad: mejor plain 50.0 % (1/2), peor parameterized 0.0 % (0/1).
- Errores de formulación más frecuentes: missed_step=4.
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- Costo total 0.0697 USD; el método más caro fue single_call:structured con 0.0697 USD (100 %).
- Métodos con tools: 1.00 tool calls medias por item (máximo 1).

6. Archivos

Rutas absolutas bajo k3_control_single_call. report.rescored.json (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
↳ /power-flow-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call/report.json [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
↳ ds-Code-v2/power-flow-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
↳ ds-Code-v2/power-flow-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
↳ s-Code-v2/power-flow-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call/scoreboard.rescored.json
scoreboard_per_case.rescored.json:
↳ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/
↳ ower-flow-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call/scoreboard_per_case.rescored.json
traces (5 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/
↳ rtGrids-Code-v2/power-flow-agent/benchmarks/results_validation_k_sweep/gpt-5.4/k3_control_single_call/traces

```